

① Given $A = \begin{pmatrix} -15 & 1 & 11 \\ \underline{20} & 32 & -23 \\ 0 & -15 & 35 \end{pmatrix}$ 9 17

$$u = \begin{pmatrix} -15 \\ 20 \end{pmatrix} \quad c = \frac{u_1}{\|u\|_2} \quad , \quad s = \frac{u_2}{\|u\|_2}$$

$$\|u\|_2 = \sqrt{15^2 + 20^2} = 25, \quad = -\frac{15}{25} \quad , \quad = \frac{20}{25}$$

$$c = -\frac{3}{5} \quad , \quad s = \frac{4}{5}$$

$$\rightarrow c = \frac{3}{5} \quad , \quad s = -\frac{4}{5}$$

$$\tilde{Q}_{21} = \begin{pmatrix} c & s \\ -s & c \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$$

$$Q_{21} = \begin{pmatrix} \tilde{Q}_{21} & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & -4 & 0 \\ 4 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

$$Q_{21} A = \frac{1}{5} \begin{pmatrix} \underline{3} & \underline{-4} & 0 \\ \underline{4} & \underline{3} & 0 \\ \underline{0} & \underline{0} & \underline{5} \end{pmatrix} \begin{pmatrix} -\underline{15} & 1 & 11 \\ \underline{20} & 32 & -23 \\ \underline{0} & -15 & 35 \end{pmatrix}$$

$$= \frac{1}{5} \begin{pmatrix} -125 & -125 & 125 \\ 0 & 100 & -25 \\ 0 & -75 & 175 \end{pmatrix}$$

$$= \begin{pmatrix} \underline{-25} & -25 & +25 \\ \underline{0} & 20 & -5 \\ \underline{0} & -15 & 35 \end{pmatrix}$$

$$s = -\frac{4}{5}$$

$$u = \begin{pmatrix} -25 \\ 0 \end{pmatrix} \Rightarrow \|u\|_2 = 25, \quad c = \frac{u_1}{\|u\|_2} = -1$$

$$\underline{s = 0}$$

$$c \stackrel{c \leq 0}{=} 1, \quad s = 0$$

$$Q_{31} = \begin{pmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{pmatrix} = I$$

$$Q_{31} Q_{21} A = Q_{21} A = \left(\begin{array}{c|cc} -25 & -25 & 25 \\ 0 & 20 & -5 \\ 0 & -15 & 35 \end{array} \right)$$

$$u = \begin{pmatrix} 20 \\ -15 \end{pmatrix}, \quad \|u\|_2 = \sqrt{20^2 + 15^2} = 25$$

$$c = \frac{u_1}{\|u\|_2} = \frac{20}{25} = \frac{4}{5} > 0$$

$$s = \frac{u_2}{\|u\|_2} = -\frac{15}{25} = -\frac{3}{5}$$

$$Q_{32} = \frac{1}{5} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -3 \\ 0 & 3 & 4 \end{pmatrix}$$

$$Q_{32} (Q_{31} Q_{21} A)$$

$$= \frac{1}{5} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -3 \\ 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} -25 & -25 & 25 \\ 0 & 20 & -5 \\ 0 & -15 & 35 \end{pmatrix}$$

$$= \dots = \left(\begin{array}{c|cc} -25 & -25 & 25 \\ 0 & 25 & -25 \\ 0 & 0 & 25 \end{array} \right) = R$$

Spencer

$$\begin{array}{r} -25 \quad -25 \quad 25 \\ \hline -\frac{4}{5} \quad \left| \quad 25 \quad -25 \right. \\ \hline \quad \quad -\frac{3}{5} \quad \left| \quad 25 \right. \\ \hline \quad \quad \quad 0 \end{array}$$

↗

$$s, c = \sqrt{1-s^2}$$